

## Inverses from Graphs and Tables

**TEACHER KEY** | Algebra II | Unit 1 | Day 10 | TEK A2.2.B

**Objective:** I can find the inverse of a relation from a table or a graph, use  $f^{-1}(x)$  notation, and recognize that a function and its inverse are reflections across the line  $y = x$ .

### Vocabulary

**Inverse** — the relation that **undoes** the original.

**Notation** — we write  **$f^{-1}(x)$**  for the inverse of  $f(x)$ . Careful: this does **not** mean  $1/f(x)$ .

**In a table** — to find the inverse, **swap x and y** in every pair.

**On a graph** — a function and its inverse are reflections across the line  **$y = x$** .

### Part 1 — Inverses from Ordered Pairs

Swap the coordinates in each pair.

**Example 1.**  $f = \{ (-4, -6), (0, -1), (2, 4), (3, -7), (7, 7) \}$

$f^{-1} = \{ \textcolor{red}{(-6, -4)}, \textcolor{red}{(-1, 0)}, \textcolor{red}{(4, 2)}, \textcolor{red}{(-7, 3)}, \textcolor{red}{(7, 7)} \}$

Domain of  $f = \{ \textcolor{red}{-4, 0, 2, 3, 7} \}$  which becomes the **range** of  $f^{-1}$ .

**You Try 1.**  $g = \{ (1, 5), (2, 8), (3, 11) \}$        $g^{-1} = \{ \textcolor{red}{(5, 1)}, \textcolor{red}{(8, 2)}, \textcolor{red}{(11, 3)} \}$

### Part 2 — Inverses from a Table

Fill in the inverse table by trading the rows.

|             |    |    |   |    |   |
|-------------|----|----|---|----|---|
| <b>x</b>    | -4 | 0  | 2 | 3  | 7 |
| <b>f(x)</b> | -6 | -1 | 4 | -7 | 7 |

Now the inverse:

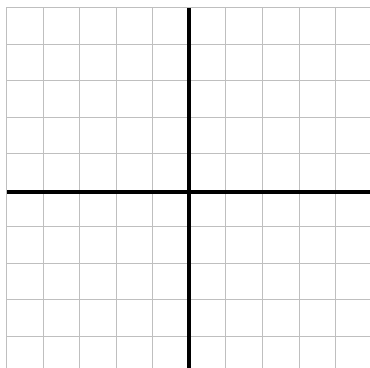
|                               |    |    |   |    |   |
|-------------------------------|----|----|---|----|---|
| <b>x</b>                      | -6 | -1 | 4 | -7 | 7 |
| <b><math>f^{-1}(x)</math></b> | -4 | 0  | 2 | 3  | 7 |

The two tables are the same information, just **read the other way**.

### Part 3 — Graph a Function and Its Inverse

Each square is 1 unit. Plot  $f$ , then plot  $f^{-1}$ , then draw the line  $y = x$  through the origin at  $45^\circ$ .

**A.** Plot  $f = \{ (-4, -2), (0, 0), (1, 1), (2, 4) \}$  and its inverse.



**B.** What do you notice?

The two graphs are mirror images across the line  **$y = x$**

Which points stay put under the swap?  
**points already on  $y = x$ , like (0,0) and (1,1)**

If  $(a, b)$  is on  $f$ , then on  $f^{-1}$  you find  
 **$(b, a)$**

$f^{-1} = \{ \textcolor{red}{(-2, -4)}, \textcolor{red}{(0, 0)}, \textcolor{red}{(1, 1)}, \textcolor{red}{(4, 2)} \}$

## Part 4 — Domain and Range Trade Places

For any relation: domain of  $f$  = **range** of  $f^{-1}$ , and range of  $f$  = **domain** of  $f^{-1}$ .

| Function          | Domain of $f$ | Range of $f$  | Domain of $f^{-1}$              | Range of $f^{-1}$               |
|-------------------|---------------|---------------|---------------------------------|---------------------------------|
| $f(x) = x^3$      | all reals     | all reals     | <b>all reals</b>                | <b>all reals</b>                |
| $f(x) = 2^x$      | all reals     | $(0, \infty)$ | <b><math>(0, \infty)</math></b> | <b>all reals</b>                |
| $f(x) = \sqrt{x}$ | $[0, \infty)$ | $[0, \infty)$ | <b><math>[0, \infty)</math></b> | <b><math>[0, \infty)</math></b> |

Row 2 explains Day 7: the exponential and the **logarithm** are inverses, which is why their domains and ranges swapped.

## Part 5 — Quick Check

- Does  $f^{-1}(x)$  mean 1 divided by  $f(x)$ ? **no**
- If  $(3, -5)$  is on the graph of  $f$ , name a point on  $f^{-1}$ .  **$(-5, 3)$**
- A graph and its inverse are reflections across which line?  **$y = x$**

## Practice

Find each inverse. Show your work where there is room.

- $p = \{ (0, 3), (1, 5), (2, 7) \}$      $p^{-1} = \{ \mathbf{(3, 0), (5, 1), (7, 2)} \}$
- Domain of  $p = \mathbf{\{0, 1, 2\}}$     Range of  $p^{-1} = \mathbf{\{0, 1, 2\}}$
- A table gives  $f$ :  $(2, 6), (4, 6), (5, 1)$ . Write  $f^{-1}$ , then decide whether it is a function.

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$f^{-1} = \{ \mathbf{(6, 2), (6, 4), (1, 5)} \}$     A function? **no - 6 has two outputs**

- That is a preview of Day 12. What went wrong in item 3?

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- If a graph contains  $(-3, 7)$ , name the matching point on its inverse.  **$(7, -3)$**

## Exit Ticket

A function  $h$  has the table below.

$h$ :  $(1, 4), (2, 9), (5, 0)$      $h^{-1} = \{ \mathbf{(4, 1), (9, 2), (0, 5)} \}$

Domain of  $h^{-1}$ :  **$\{0, 4, 9\}$**     Range of  $h^{-1}$ :  **$\{1, 2, 5\}$**